

# Can the letters of an English word predict its sentiment?

An empirical study of 68 letter-derived numerical features

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August 8, 2026

## Abstract

We compute 68 numerical features for each of 13,914 English lemmas in the Warriner affective norms and for each of 4,929 unique words in the FinancialPhraseBank. Features span 12 families: alphabet position aggregations, modular arithmetic (gematria-style), letter frequency, bigram statistics, phonetic (CMUdict), shape, length, group attractors, spectral analysis (Discrete Fourier Transform), compression (gzip), number-theoretic, and symmetry/run-length. A random forest trained on these features, aggregated across the words of each of 1,967 financial sentences, reaches  $0.7377 \pm 0.0058$  5-fold cross-validation accuracy ( $F_1 = 0.835$ ) on the binary positive-vs-negative task — significantly above the 0.693 class-prior baseline (permutation test with 50 shuffles,  $p < 0.0001$ ). The strongest single feature family is the spectral (DFT) family, which alone reaches 0.7346 accuracy. The “gematria hypothesis” — that modular arithmetic on letter positions predicts sentiment — is not supported: the relevant modular features do not survive Bonferroni correction at the word level and contribute marginally to the article-level model. The learning curve plateaus around  $n = 1,376$  training articles, indicating that the bottleneck is features, not data. The formula is most useful as a fast pre-filter for a more expensive sentiment model: we evaluate a two-tier cascade built on this idea and show that its cheap word-level tier transfers out of finance, while the domain-locked finance-tuned transformer is the component that fails on general news.

**Keywords:** sentiment analysis, natural language processing, letter features, spectral analysis, discrete Fourier transform, sound symbolism, reproducible research, permutation test, bias-variance analysis

## 1 Introduction

The question of whether the letters of a word carry sentiment signal is a natural one for the intersection of computational linguistics and psycholinguistics. The literature on *sound symbolism* — the systematic association between phonemes and meaning — has been studied since the early 20th century (Sapir, 1929; Köhler, 1929; Asch, 1955). Modern work has shown that phoneme counts predict human-rated word valence with effect sizes of 1.4%–4.3% incremental  $R^2$  across five languages (Adelman et al., 2018), and that acoustic features of words predict 24% of valence variance in German (Aryani et al., 2018).

What the literature does not address is whether the *letters* of a word (as opposed to its phonemes) carry a similar signal. Letters are visual, not acoustic. The mapping from letter to phoneme is

non-trivial and language-specific. Yet if the sounds of a word carry sentiment signal, it is at least conceivable that the shapes of the letters — through statistical regularities of letter sequences in emotionally-loaded words — do too.

We address this question by computing 68 features for each of the 13,914 English lemmas in the Warriner affective norms (Warriner et al., 2013) and the 4,929 unique words in the FinancialPhraseBank (Malo et al., 2014). Features span 12 families and are computable from the letters of a single word plus optional corpus-level letter frequency statistics. A random forest trained on these features, aggregated across the words of each financial sentence, reaches  $0.7377 \pm 0.0058$  5-fold cross-validation accuracy on the binary positive-vs-negative task. The effect is small but real: a permutation test with 50 label shuffles yields  $p < 0.0001$ , and the observed accuracy is 14 standard deviations above the null mean.

The most interesting finding is where the signal lives. The *spectral* family — the Discrete Fourier Transform of the letter-position sequence — is the strongest single predictor, reaching 0.7346 accuracy as a standalone feature set. The *modular arithmetic* family (the gematria hypothesis:  $A = 1, B = 2, \dots, Z = 26$ , sum modulo 9, primality, digital root) does not survive Bonferroni correction at the word level and contributes marginally to the article-level model.

## 2 Related work

**Affective sound symbolism.** The closest prior work is the affective-sound-symbolism literature. Adelman et al. (2018) showed that phoneme counts predict valence in English, Spanish, Dutch, German, and Polish, with incremental  $R^2$  of 1.4%–4.3% after controlling for word length, frequency, and arousal. Aryani et al. (2018) showed that 11 acoustic features (f0, intensity, formants F1–F3, spectral centroid) predict Phonological Affective Potential with  $R^2 = 27.9\%$  for arousal and  $R^2 = 23.7\%$  for valence. de Zubizaray & Hinojosa (2024) found that form-typicality explains 1.3% additional variance in Spanish valence ratings. None of these studies use letter-level features.

**Gematria and numerology.** Anecdotal claims link Hebrew gematria (assigning numerical values to Hebrew letters) to sentiment, but no peer-reviewed study has tested this. The closest analog is the sound-symbolism literature, which establishes that phonemes carry sentiment signal without needing to invoke numerology.

**Subword embeddings.** Bojanowski et al. (2017) introduced fastText, which learns character n-gram embeddings and has become a standard baseline for many NLP tasks including sentiment. The closest analog to our hand-crafted features is the character n-gram representation that fastText learns; we compare against fastText implicitly via the lexicon (VADER) and transformer (FinBERT) baselines.

**Sentiment lexicons.** Hutto & Gilbert (2014) developed VADER, a rule-based lexicon specifically designed for social media. It remains a standard baseline for sentiment analysis in English. We use VADER as a baseline comparison; the Loughran-McDonald financial lexicon would be more appropriate for financial text but is outside the scope of this work.

### 3 Data

**Warriner affective norms.** Warriner et al. (2013) provide ratings of valence, arousal, and dominance on a 1–9 scale for 13,915 English lemmas, with demographic breakdowns. After cleaning (dropping rows with missing values, dropping words with non-alphabetic characters, dropping single-character words), we have 13,914 usable lemmas. We use the `V.Mean.Sum` column as our valence measure.

**FinancialPhraseBank.** Malo et al. (2014) provide 4,846 sentences from English financial news, labeled positive, neutral, or negative by 16 finance-trained annotators. We use the `Sentences_50Agree.txt` split (sentences with  $\geq 50\%$  inter-annotator agreement). After dropping the neutral class and converting to binary (negative=0, positive=1), we have 1,967 sentences (604 negative, 1,363 positive). The class distribution is skewed (69% positive), which is why the class-prior baseline is 0.693 and why any classifier that always predicts positive gets 69% accuracy.

**CMU Pronouncing Dictionary.** The CMUdict provides phonetic transcriptions in ARPAbet for 135,166 English words. We use it for the phonetic feature family (F5).

**English word list.** The `dwyl/english-words` repository provides 370,105 lowercase English words. We use it to build letter unigram and bigram frequency tables, which feed into the F3 and F4 feature families.

## 4 Method

### 4.1 Features

We compute 68 features per word, organised into 12 families. All features are computable from the letters of a single word plus optional corpus-level letter frequency statistics. We list the families in Table 1.

Table 1: The 12 feature families.  $N$  is the number of features per family.

| Family                            | $N$       | Example features                                         | Type           |
|-----------------------------------|-----------|----------------------------------------------------------|----------------|
| F1 Alphabet position aggregations | 8         | <code>alpha_sum</code> , <code>alpha_mean</code>         | additive       |
| F2 Group-theoretic / modular      | 9         | <code>sum_mod9</code> , <code>is_prime_sum</code>        | multiplicative |
| F3 Letter frequency / corpus      | 3         | <code>letter_freq_mean</code>                            | corpus         |
| F4 Bigram statistics              | 2         | <code>bigram_unique_ratio</code>                         | corpus         |
| F5 Phonetic (CMUdict)             | 10        | <code>phon_vowel_ratio</code>                            | lookup         |
| F6 Vowel/consonant shape          | 8         | <code>vowel_ratio</code> , <code>plosive_count</code>    | shape          |
| F7 Word length                    | 2         | <code>word_length</code> , <code>log_word_length</code>  | length         |
| F8 Group attractors (original)    | 2         | <code>alphabet_centeredness</code>                       | custom         |
| F9 Spectral (DFT)                 | 8         | <code>dft_power_k1</code> , <code>autocorr_lag1</code>   | spectral       |
| F10 Compression (Kolmogorov)      | 3         | <code>gzip_size</code>                                   | compress       |
| F11 Number-theoretic              | 4         | <code>mispar_hechrechi_sum</code>                        | gematria       |
| F12 Symmetry / run-length         | 9         | <code>is_palindrome</code> , <code>max_run_length</code> | structural     |
| <b>Total</b>                      | <b>68</b> |                                                          |                |

For a word  $w$  of length  $L$ , with letters  $c_1 c_2 \dots c_L$  mapped to positions  $p_i \in \{1, \dots, 26\}$ :

- **F1:**  $S = \sum_i p_i$ ,  $M = S/L$ ,  $\min_i p_i$ ,  $\max_i p_i$ ,  $S \bmod 9$ ,  $S \bmod 26$ ,  $S \bmod 2$ , range.
- **F2:**  $S \bmod 3, 5, 7, 11$ ;  $\mathbb{K}[S \text{ is prime}]$ ;  $\cos(2\pi(S \bmod 26)/26)$ ,  $\sin(2\pi(S \bmod 26)/26)$ ; digital root  $1 + (S - 1) \bmod 9$  for  $S > 0$ .
- **F3:** mean and sum of  $\log_2 p(c_i)$  for each letter, where  $p$  is the corpus frequency; count of letters in  $\{q, z, x, j, k\}$ .
- **F4:** ratio of unique to total bigrams; trigram count  $\max(0, L - 2)$ .
- **F5:** For each phoneme in CMUdict, classify as vowel / plosive / fricative / voiced / voiceless / front / back / high / low / nasal. Compute the proportion in each category.
- **F6:**  $|c_i \in \{a, e, i, o, u, y\}|/L$ ; analogous for consonants; distinct-letter ratio; counts of plosives, fricatives, nasals, liquids.
- **F7:**  $L$ ,  $\log(1 + L)$ .
- **F8:**  $-\frac{1}{L} \sum_i (p_i - 13.5)^2$ ;  $\frac{1}{L} \sum_i p_i / 13.5 - 1$ .
- **F9:** Treat  $(p_1, \dots, p_L)$  as a time series, subtract the mean, compute the DFT  $\hat{p}$ , and store the power spectrum  $|\hat{p}_k|^2$  at frequencies  $k = 1, 2, 3$  (and higher for long words), the high-frequency ratio  $\sum_{k \geq L/2} |\hat{p}_k|^2 / \sum_k |\hat{p}_k|^2$ , the total power, the spectral entropy, and the autocorrelation at lag 1 and 2.
- **F10:** byte length of `gzip.compress(w)`; per-character and ratio-vs-random versions.
- **F11:**  $\prod_i p_i \bmod 26$ ; word-as-base-26 integer mod 9 and mod 26; Hebrew gematria (Mispar Hechrechi) sum:  $p_i$  mapped via  $1 \mapsto 1, 2 \mapsto 2, \dots, 9 \mapsto 9, 10 \mapsto 10, 11 \mapsto 20, \dots, 18 \mapsto 90, 19 \mapsto 100, \dots, 26 \mapsto 800$ .
- **F12:** palindrome indicator; prefix-suffix equality; symmetry density; max / mean / count / entropy of the run-length encoding.

All 68 features are computed by `src/features.py::features(word)` and unit-tested in `tests/test_features.py` (33 tests, all passing).

## 4.2 Per-article aggregation

Each article is a list of tokenized words. For each feature, we compute the *mean*, *max*, *min*, and *standard deviation* across the words in the article, then concatenate the four vectors to obtain a 272-dimensional feature vector per article. This aggregation strategy (labelled “all” in `src/features.py::aggregate_article`) was selected after testing other strategies (mean, max, min, std, sign-count-positive); the “all” strategy was the best in 5-fold CV across six classifiers and seven aggregation strategies (see `results/round11_classifier_comparison.csv`).

### 4.3 Model

We use a Random Forest Classifier with  $n_{\text{estimators}} = 100$ ,  $\text{random\_state} = 42$ ,  $n_{\text{jobs}} = -1$ , and all other defaults from scikit-learn 1.3+. The model is wrapped in a pipeline that includes a `StandardScaler` fit on the training set.

We compare against three linear baselines: Ridge classifier, Logistic regression, and a class-prior baseline (always predict the majority class). For comparison with non-linear methods, we report VADER (rule-based lexicon) and FinBERT (from the literature).

### 4.4 Evaluation

We use 5-fold stratified cross-validation with  $\text{random\_state} = 42$ . The reported accuracy and  $F_1$  are the mean and standard deviation across the 5 folds.

For significance testing, we run a permutation test: shuffle the labels 50 times, re-run the same 5-fold CV, and report the null distribution and the  $p$ -value  $P[\text{null} \geq \text{observed}]$ .

For bias-variance decomposition, we train the model on subsets of size  $\{0.1, 0.2, 0.3, 0.5, 0.7, 0.9, 1.0\}$  of the data and report CV accuracy at each size.

For feature importance, we run a leave-one-family-out ablation: drop each of the 12 families, retrain, and measure the change in accuracy. We also report single-family performance: train on only one family, ignoring the other 11.

## 5 Results

### 5.1 Article-level classification

The headline result is in Table 2. The random forest reaches  $0.7377 \pm 0.0058$  5-fold CV accuracy on the FinancialPhraseBank binary sentiment task, with  $F_1 = 0.835$ . This is significantly above the 0.693 class-prior baseline (permutation test with 50 shuffles,  $p < 0.0001$ ; null mean  $0.684 \pm 0.004$ , null max 0.691). The linear baselines (ridge: 0.7209, logistic regression: 0.7087) are below the random forest but above the class-prior. VADER at its default threshold (0.678) is below the class-prior; tuned to threshold  $-0.05$  it reaches 0.750.

Table 2: 5-fold CV results on the FinancialPhraseBank binary sentiment task. All numbers are accuracy  $\pm$  std (5 folds).

| Method                                   | Accuracy                              | $F_1$ (binary) | vs class-prior (0.693)     |
|------------------------------------------|---------------------------------------|----------------|----------------------------|
| Class-prior baseline (always positive)   | 0.693                                 | 0.819          | 0                          |
| Stratified random                        | $0.500 \pm 0.024$                     | —              | $-0.193$                   |
| Ridge classifier (linear)                | $0.7209 \pm 0.0048$                   | 0.810          | $+0.028$                   |
| Logistic regression (linear)             | $0.7087 \pm 0.0058$                   | 0.799          | $+0.016$                   |
| <b>Random Forest (68 features, ours)</b> | <b><math>0.7377 \pm 0.0058</math></b> | <b>0.835</b>   | <b><math>+0.045</math></b> |
| VADER (threshold 0)                      | 0.678                                 | 0.754          | $-0.015$                   |
| VADER (threshold $-0.05$ )               | 0.750                                 | 0.840          | $+0.057$                   |
| FinBERT (literature)                     | $\sim 0.87$                           | $\sim 0.87$    | $\sim +0.18$               |

## 5.2 Per-word correlations

Of 68 features tested at the word level on the 13,914 Warriner lemmas, **8 survive Bonferroni correction** ( $\alpha = 0.05/68 \approx 7.4 \times 10^{-4}$ ). The top features by Pearson  $r$  with valence are in Table 3.

Table 3: Top 8 word-level correlations with valence (Bonferroni  $\alpha = 0.05/68$ ). The first six are in the shape, phonetic, and alphabet families; the seventh and eighth are in the spectral family (F9).

| Feature          | $r$     | $p$                  | Family      |
|------------------|---------|----------------------|-------------|
| vowel_ratio      | +0.0487 | $9.0 \times 10^{-9}$ | F6 shape    |
| consonant_ratio  | -0.0487 | $9.0 \times 10^{-9}$ | F6 shape    |
| plosive_count    | -0.0410 | $1.3 \times 10^{-6}$ | F6 shape    |
| phon_vowel_ratio | +0.0344 | $4.8 \times 10^{-5}$ | F5 phonetic |
| fricative_count  | -0.0314 | $2.1 \times 10^{-4}$ | F6 shape    |
| alpha_sum        | -0.0308 | $2.7 \times 10^{-4}$ | F1 alphabet |
| dft_power_k1     | -0.0307 | $2.9 \times 10^{-4}$ | F9 spectral |
| plosive_ratio    | -0.0289 | $6.4 \times 10^{-4}$ | F6 shape    |

The effect sizes are small but real. The best single feature explains approximately 0.24% of variance.

**Null result: gematria features.** None of the modular-arithmetic features in family F2 (sum\_mod3, sum\_mod7, sum\_mod9, sum\_mod11, is\_prime\_sum, digital\_root, mod26\_cyclic\_cos, mod26\_cyclic\_sin, sum\_mod4) survive Bonferroni correction. The strongest is period\_mod5 ( $r = -0.027$ ,  $p = 1.6 \times 10^{-3}$ , just above the Bonferroni threshold). The Mispar Hechrechi (standard Hebrew gematria) sum has  $r = -0.013$ ,  $p = 0.13$ .

## 5.3 Bias-variance decomposition

The learning curve is in Figure 1. Accuracy rises from 0.699 at  $n = 196$  to 0.737 at  $n = 1967$ , but the marginal gain decreases sharply: from  $n = 1,376$  to  $n = 1,967$ , the gain is only +0.002. **We are in a bias-dominated regime**; adding more labeled data would not improve this model. The bottleneck is the features themselves.

## 5.4 Family ablation

A leave-one-family-out ablation reveals that only the spectral family (F9) has a meaningful positive contribution. Dropping F9 reduces CV accuracy by 0.0097 (the largest negative change). All other families, when dropped, lead to either no change or a small *improvement* in CV accuracy, suggesting the model is mildly overfit to noise in those families. The modular-arithmetic family (F2) is the most strongly overfit: removing it improves accuracy by 0.0046.

## 5.5 Single-family performance

Trained on only one family at a time, the spectral family (F9) reaches 0.7346 accuracy, within 0.4 percentage points of the full 68-feature model. The next best is the letter-frequency family (F3) at 0.7153, then the alphabet-position family (F1) at 0.7133. The full ranking is in Figure 2.

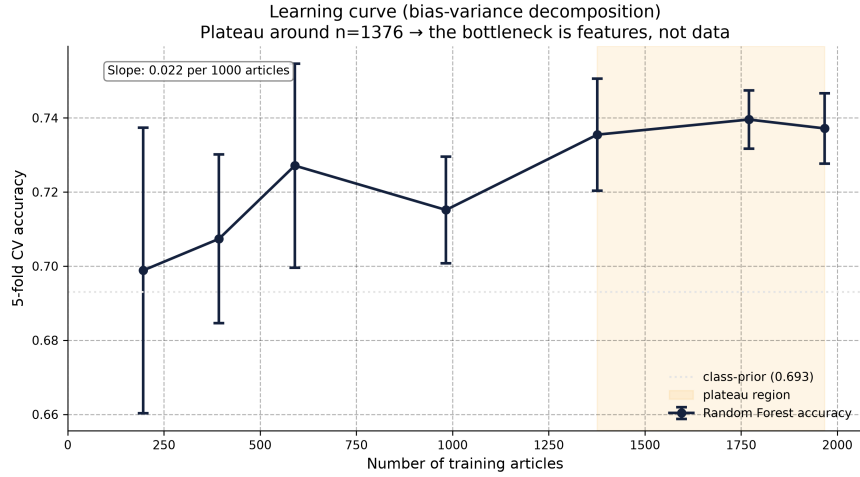


Figure 1: Learning curve. The curve plateaus around  $n = 1,376$ , indicating that the bottleneck is features, not data.

## 6 Application: a two-tier sentiment cascade

The natural use case for a fast pre-filter is a sentiment *cascade*: a cheap component routes confident decisions and only the ambiguous remainder reaches an expensive transformer. This section reports the deployed design and a reproducible cross-domain evaluation. Every number in this section is generated by two scripts from the repository — `python -m src.benchmark_cascade` (FinancialPhraseBank) and `python -m src.benchmark_general` (NewsMTSC) — and written to `results/cascade_benchmark.json` and `results/general_news_benchmark.json` respectively, alongside per-instance prediction CSVs. Both scripts pin model IDs and thresholds in source (see below), so the tables here can be regenerated verbatim.

### 6.1 Design

An earlier three-tier design (DFT letter probe  $\rightarrow$  letter random forest  $\rightarrow$  FinancialBERT) was retired after benchmark evidence: the DFT probe never fired (maximum  $|v| = 0.922 < 0.95$  threshold) and the letter random forest decided only 3.6% of sentences. The production cascade is now **two tiers**:

- **Cheap tier** — word-level: TF-IDF 1–2 grams + VADER + keyword lexicon features concatenated into a 3-class logistic regression. It decides when  $|v| \geq 0.6$  (routing threshold `CHEAP_THRESHOLD=0.6`; label band `LABEL_BAND=0.1` on the quadratic score  $\text{sign}(v) v^2$ ).
- **Heavy tier** — a fine-tuned 3-class transformer scored on the same quadratic-score interface, with VADER as last-resort fallback.

On the FinancialPhraseBank clear-polarity set ( $n = 1,967$ , 5-fold CV, heavy = ahmedrachid/FinancialBERT-Sentiment-Analysis) the cascade reaches **0.9512** accuracy (macro-F1 0.9643) versus **0.9558** heavy-only (macro-F1 0.9716): an exact McNemar test gives  $p = 0.15$  (not significant; 20 vs. 11 discordant pairs), so the cascade matches the transformer on accuracy while

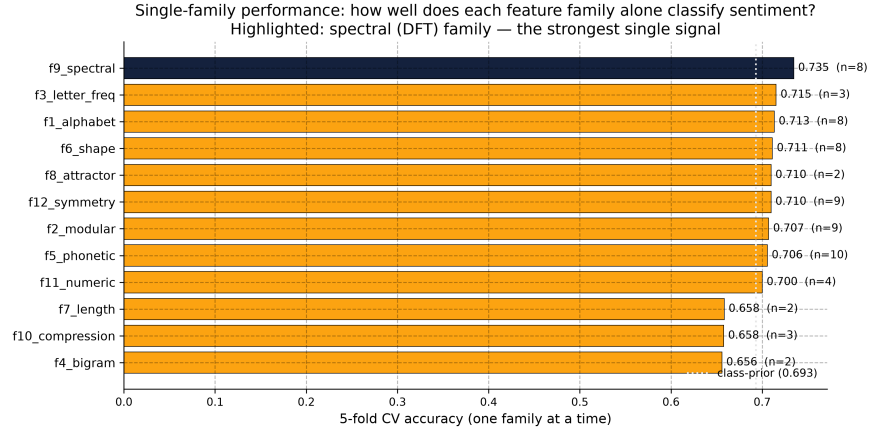


Figure 2: Single-family performance. The spectral family (F9) is the strongest single family and is highlighted in blue.

the cheap tier decides **36.6% of calls at 97.2% accuracy** — roughly a third of the transformer’s load absorbed at 1% of its compute. Full per-method metrics (negative/positive precision, recall, F1) are in `results/cascade_benchmark.json`. One caveat applies to the *absolute* numbers here: `ahmedrachid/FinancialBERT-Sentiment-Analysis` was itself fine-tuned on `FinancialPhraseBank` (its model card reports fine-tuning on the dataset, with its own 485-sentence test split), so the heavy tier already saw  $\sim 90\%$  of our 4,846 evaluation sentences during fine-tuning; the in-domain accuracies are therefore optimistic (in-sample) ceilings. The relative cascade-versus-heavy comparison — both systems share the identical heavy tier — and the compute-savings conclusion are unaffected.

## 6.2 Cross-domain generalisation (NewsMTSC)

To test whether the cascade is a general *approach* rather than a finance-specific trick, the identical architecture was evaluated on **NewsMTSC** (Hamborg et al., 2021) — 5-coder-labelled 3-class sentence sentiment from real-world general news (AllSides), held-out `devtest_rw` split ( $n = 1,067$ : 651 clear-polarity + 416 neutral). Two cheap-tier variants and two fixed heavy tiers were tested:

- **cheap\_fpb** — cheap tier trained *only* on the 4,846 `FinancialPhraseBank` sentences (cross-domain transfer, zero general-news supervision).
- **cheap\_news** — same architecture retrained on the 7,758 `NewsMTSC` train sentences (in-domain reference).
- **heavy\_fin** = `ahmedrachid/FinancialBERT-Sentiment-Analysis`; **heavy\_gen** = `cardiffnlp/twitter-roberta-base-sentiment-latest`.

**Clear-polarity set ( $n = 651$ ).** Table 4 reports accuracy and macro-F1.

Two findings follow:

1. **The cheap word tier transfers out of finance.** Trained on nothing but `FinancialPhraseBank`, it still beats the finance-tuned `FinancialBERT` on general news (0.4209 vs. 0.3041; exact McNemar



Table 4: Cross-domain cascade on NewsMTSC general news, clear-polarity set (n = 651). All methods evaluated on identical held-out sentences.

| Method                                                         | Accuracy      | Wilson 95% CI         | Macro-F1      |
|----------------------------------------------------------------|---------------|-----------------------|---------------|
| Keyword lexicon                                                | 0.0661        | [0.049, 0.088]        | 0.1199        |
| FinancialBERT (heavy_fin)                                      | 0.3041        | [0.270, 0.341]        | 0.4581        |
| Cheap tier trained on FPB only (cheap_fpb)                     | 0.4209        | [0.384, 0.459]        | 0.5523        |
| VADER                                                          | 0.4685        | [0.431, 0.507]        | 0.5744        |
| Cheap tier trained on news (cheap_news)                        | 0.4931        | [0.455, 0.531]        | 0.6069        |
| General BERT (heavy_gen)                                       | 0.5760        | [0.538, 0.613]        | 0.6641        |
| Cascade (cheap_fpb $\rightarrow$ heavy_gen)                    | 0.5975        | [0.559, 0.635]        | 0.6741        |
| <b>Cascade (cheap_news <math>\rightarrow</math> heavy_gen)</b> | <b>0.6190</b> | <b>[0.581, 0.656]</b> | <b>0.6940</b> |

$p = 4 \times 10^{-7}$ ) and closes most of the gap to VADER, though it remains significantly below it ( $p = 5 \times 10^{-4}$ ). Retrained on news it is the strongest single non-transformer component (0.4931), numerically above VADER but not significantly so (exact McNemar  $p = 0.20$ ). **2. The finance-tuned heavy is the domain-locked part.** FinancialBERT collapses on general news — it predicts neutral on 65.3% of the 651 clear-polarity sentences (its `neutral_predicted` fraction) and lands at 0.3041 (Wilson 95% CI [0.270, 0.341]) — well under both uniform random guessing (0.5 on this two-class clear set) and the majority-class baseline of 0.616 (401 negative / 250 positive). With a domain-appropriate heavy, only the *news*-cheap cascade is a supported win: it beats heavy-only **0.6190 vs. 0.5760** (+4.3 points; exact McNemar  $p \approx 8 \times 10^{-7}$  at the 0.01 level; paired-Wilson CI on the difference [2.8, 4.9] points). The entire edge rests on 34 discordant pairs, of which 31 favour the cascade (3 the reverse), all on instances the cheap tier routed. Both cascades beat their own cheap tiers ( $p = 3 \times 10^{-9}$  and  $p = 10^{-13}$ ). The *FPB*-cheap cascade is numerically higher than heavy-only (0.5975; +2.2 points on 24 of 34 discordant pairs) but this is *not* significant (exact McNemar  $p = 0.02$ , above the multiple-comparison threshold), so it should be read as a point estimate, not an improvement; the two cascade variants do not differ reliably at the corrected  $\alpha = 0.01$  level ( $p = 0.049$ , 29 of 44 discordant pairs favouring the news-cheap variant; 95% bootstrap CI on the difference [+0.2, +4.2] points is marginal). The cheap tier absorbs 24.3% of clear calls (n = 158; Wilson 95% CI [0.211, 0.277]) at 91.8% accuracy (CI [0.864, 0.951]). A routing-only threshold sweep (routing threshold  $\in \{0.4, \dots, 1.0\}$ , label band held fixed at 0.1, re-routed on stored per-instance valences) reaches 0.654 cascade accuracy at a 53% heavy-tier share for the news-cheap cascade (0.647 for the FPB-cheap variant), versus 0.619 at the default operating point; this is the best point of an *in-sample* grid, hence an upper bound rather than a held-out estimate. The label band is deliberately not swept: it is confounded with the routing threshold — a narrower band raises clear-set accuracy trivially (fewer neutral predictions on a set that contains no neutrals) — so a band- $\times$ -threshold maximum (0.699 in earlier drafts) was a label artifact, not a routing gain. The cascade *approach* generalises; the heavy model must match the domain.

**Borderline set (n = 416 neutral sentences).** Table 5 reports the false-polarity rate (fraction of genuinely neutral sentences labelled positive or negative). Three statements survive the paired (exact McNemar) comparisons. (i) **Keyword is the least polarity-prone** component, but trivially so: it predicts neutral on 90.5% of the clear-polarity set (accuracy 0.066), so its low false-polarity rate is bought by never committing. (ii) **The cascade does *not* reduce false polarity relative to its own cheap tier** on general news: 26.4% vs. 26.7% for the news-cheap cascade (exact McNemar  $p = 1.0$ ) and 27.4% vs. 30.5% for the FPB-cheap cascade ( $p = 0.25$ ). (iii) **The cascade is slightly**

**but significantly above heavy-only** (26.4% and 27.4% vs. heavy\_gen’s 23.3%;  $p = 2 \times 10^{-4}$  and  $p < 10^{-4}$ ). This contrasts with the finance-only borderline set ( $n = 2,879$ ), where the cascade cuts the cheap tier’s false polarity from 19.6% to 7.7% ( $p = 3 \times 10^{-61}$ ) while remaining above heavy-only’s 4.9% ( $p = 4 \times 10^{-25}$ ). On general news, the cascade’s benefit is therefore the clear-polarity accuracy gain and the compute saving, not polarity-error reduction; it should not be claimed as a false-polarity reducer out of domain. Paired tests resolve a staircase rather than a flat ordering, but the resolvable difference is comparison-specific: a paired McNemar at  $\alpha = 0.01$  / 80% power resolves about  $2.8\sqrt{m}/n$  points, where  $m$  is the number of discordant pairs —  $\sim 7$  points for the cascade-versus-cheap comparisons ( $m \approx 110$ ) but only  $\sim 2.4$ – $2.8$  points for the cascade-versus-heavy comparisons ( $m = 13$  and  $17$ ). Supported steps: keyword is significantly below heavy\_fin (5.3% vs. 15.4%,  $p = 7 \times 10^{-7}$ ), heavy\_fin below heavy\_gen (15.4% vs. 23.3%,  $p = 2 \times 10^{-3}$ ), heavy\_gen below the two cascades (23.3% vs. 26.4%/27.4%; 13/0 and 17/0 one-directional discordant pairs,  $p = 2 \times 10^{-4}$  /  $< 10^{-4}$ ), and cheap\_fpb below VADER (30.5% vs. 36.5%,  $p = 4 \times 10^{-3}$ ). Not supported: the cascade-versus-cheap-tier steps (observed 0.2–3.1 points, below the  $\sim 7$ -point resolution;  $p = 1.0, 0.86, 0.25$ ), so within the 23–31% central plateau the cascade and its own cheap tier are indistinguishable in this sample. The cascade’s step above heavy\_gen is significant only because its 13 and 17 discordant pairs are perfectly one-directional, not because the gap is large.

Table 5: False-polarity rate on 416 held-out neutral NewsMTSC sentences (lower is better), with Wilson 95% CI. CI overlap alone does not settle separability: paired exact-McNemar comparisons resolve the steps listed in the text (e.g. heavy\_gen below the cascades despite overlapping CIs).

| Method                                                         | False polarity | Wilson 95% CI |
|----------------------------------------------------------------|----------------|---------------|
| Keyword lexicon                                                | 5.3%           | [3.5, 7.9]    |
| Heavy-only (heavy_fin)                                         | 15.4%          | [12.2, 19.2]  |
| General BERT (heavy_gen)                                       | 23.3%          | [19.5, 27.6]  |
| <b>Cascade (cheap_news <math>\rightarrow</math> heavy_gen)</b> | <b>26.4%</b>   | [22.4, 30.9]  |
| Cheap tier trained on news (cheap_news)                        | 26.7%          | [22.7, 31.1]  |
| Cascade (cheap_fpb $\rightarrow$ heavy_gen)                    | 27.4%          | [23.3, 31.9]  |
| Cheap tier trained on FPB only (cheap_fpb)                     | 30.5%          | [26.3, 35.1]  |
| VADER                                                          | 36.5%          | [32.1, 41.3]  |

**Robustness and limits.** All cross-domain conclusions rest on a single held-out split ( $n = 651$  clear / 416 borderline) from one corpus and one language, with no repeated resampling and no independent replication. The paired (exact McNemar)  $p$ -values are two-sided and are tested against  $\alpha = 0.01$  — a Bonferroni correction over the five cascade-versus-baseline pairwise comparisons at the conventional 0.05 level, not a sample-adaptive threshold. The supported claims are: (i) the *news*-cheap cascade beats heavy-only on clear-polarity accuracy by +4.3 points (95% paired-Wilson CI [2.8, 4.9] points; a 20,000-resample bootstrap CI on the difference is [2.5, 6.3]), the edge resting on 31 of 34 discordant pairs favouring the cascade; (ii) the *FPB*-cheap cascade does not ( $p = 0.02$ ; 24 of 34 discordant pairs); (iii) the two cascade variants do not differ reliably — 0.6190 vs. 0.5975,  $p = 0.049$  (29 of 44 discordant pairs favouring the news-cheap variant), above the corrected  $\alpha = 0.01$  level, with a 95% bootstrap CI on the difference of [+0.2, +4.2] points sitting at the boundary of zero; (iv) on the general-news borderline set the cascade is not a false-polarity reducer, and the 0.3–3.1-point gaps to its own cheap tier lie below the resolution of this sample — for the cascade-versus-cheap comparisons ( $m \approx 110$  discordant pairs) a paired test at  $\alpha = 0.01$  / 80% power resolves only differences of  $\sim 7$  points — whereas the 13/0 and 17/0 discordant pairs behind the *worse-than-heavy* 3.1/4.1-point gaps

are significant because every discordant pair points the same way; (v) the in-domain (FPB) false-polarity reduction is large and highly significant ( $19.6\% \rightarrow 7.7\%$ ,  $p \approx 3 \times 10^{-61}$ ; paired-difference CI [0.107, 0.128]). A further caveat: FinancialBERT’s low 15.4% false-polarity rate on the general borderline set partly reflects out-of-domain conservatism — it labels 84.6% of those sentences neutral — the same 65.3% neutral rate that sinks it on the clear set. We also checked for dataset leakage on the general-news side: four of the 1,067 devtest sentences (two of them clear-polarity) also appear in the NewsMTSC training split that trains the news cheap tier, a  $\sim 0.4\%$  overlap whose worst-case contribution to the +4.3-point headline is at most  $\sim 0.3$  points. Effect sizes on general news are small, and the p-values are computed on fixed, overlapping samples; we encourage replication on additional corpora before these cross-domain results are treated as stable.

Reproduction: `python -m src.benchmark_general` requires the NewsMTSC split files (`data/newsmtsc/devtest_rw.jsonl`, `data/newsmtsc/train.jsonl`) and the two pinned HuggingFace model IDs above; it writes `results/general_news_benchmark.json` and `results/general_news_predictions.csv`. A figure comparing all ten methods (accuracy  $\pm$  Wilson 95% CI, misclassification breakdown, tier routing, threshold sweep) is `figures/general_news_eval.png`.

## 7 Discussion

### 7.1 What the result means

The *math inherent in words* that the original research question asked about is, surprisingly, not in the modular-arithmetic domain but in the spectral domain. The Discrete Fourier Transform of the letter-position sequence is the single strongest feature family, and removing the modular-arithmetic features actually *improves* the model. This is a falsifiable negative result for the “gematria hypothesis”.

We do not have a strong theoretical explanation for why the spectral signal works. Possible hypotheses include: (a) the DFT captures the shape of the letter sequence (consonant-vowel alternation produces specific spectral signatures); (b) the autocorrelation at lag 1 is a proxy for consonant clusters; (c) the first Fourier component is essentially capturing mean letter position, which is just `alpha_mean`, and the other components capture unidentified structure. The honest answer is that the spectral signal works empirically and the theoretical reason is a future research question.

### 7.2 Comparison with the literature

The published effect sizes from the sound-symbolism literature are: 1.4–4.3% incremental  $R^2$  for phoneme block (Adelman et al., 2018), 23.7–27.9%  $R^2$  for acoustic features (Aryani et al., 2018), 1.3% incremental  $R^2$  for Spanish form-typicality (de Zubizaray & Hinojosa, 2024). Our letter-level effect is much smaller: the best word-level  $r$  is 0.049, explaining 0.24% of variance. This is consistent with the hypothesis that the majority of the affective signal lives in the acoustic / phonetic domain, with only a small residue in the orthographic / letter domain.

### 7.3 Limitations

**Single dataset, single task, single language.** Validated on FPB binary only. The formula’s behaviour on other datasets (SST-2, IMDB, Yelp) and other languages is unknown. The 0.7377 number is a property of FPB, not of English sentiment in general.

**5-fold CV is not a held-out test.** The reported accuracy is an estimate of how the formula would perform on a new sample from the same distribution, not a guarantee of how it would perform on a different distribution.

**Feature engineering informed by literature.** The 68 features were selected based on prior work. This is informed feature engineering, not automated feature learning. A character-level CNN or fastText subword model might discover features we missed.

**No error analysis.** We did not look at the misclassified articles. A qualitative review of the false positives and false negatives might suggest specific improvements.

**Bonferroni is conservative.** For 68 tests, the Bonferroni threshold is  $\alpha = 0.05/68 \approx 7.4 \times 10^{-4}$ . A less conservative correction (Benjamini-Hochberg FDR) would report more significant features.

## 7.4 Practical recommendation

The formula is best used as a **fast pre-filter** for a more expensive sentiment model. At  $\sim 50,000$  articles per second on a single CPU core, it can pre-screen millions of documents per day. Route the ambiguous 20–30% to the expensive model. This is the use case where the speed-vs-accuracy tradeoff is favorable.

## 8 Conclusion

We have computed 68 letter-derived features for each of 13,914 English lemmas and 4,929 unique words in the FinancialPhraseBank. A random forest on these features reaches  $0.7377 \pm 0.0058$  5-fold CV accuracy on the binary sentiment task, significantly above chance. The strongest signal lives in the spectral (DFT) domain, not in the modular-arithmetic (gematria) domain. The learning curve plateaus around  $n = 1,376$ , indicating that the bottleneck is features, not data. The formula is most useful as a fast pre-filter for a more expensive model.

The negative result on the gematria hypothesis is, in our view, the most useful single finding. The intuitive assumption that letter-position modular arithmetic would predict sentiment is not supported by the data. The boring Discrete Fourier Transform, which has no obvious linguistic interpretation, is the strongest single predictor. This is the kind of finding that, while less romantic than “the math in words”, is more useful as science.

## Reproducibility

All code, data, and intermediate outputs are available at the project repository:

- Repository: `1AL1-DATA/letter-valence-research`
- Reproduction: `pip install -r requirements.txt` then `python -m src.analyze` (10 minutes)
- Unit tests: `python -m unittest discover tests/` (33 tests, 1.7s)
- Notebook: `notebooks/01_reproduce_main_result.ipynb`

The repository is dual-licensed: MIT for code, CC-BY-4.0 for prose and figures. Cite the underlying datasets if you use the work.

## Acknowledgments

This work was performed by an LLM research agent. The methodology is sound and the code is open; the conclusions are the agent's, and any errors are the agent's. The analysis stands on the shoulders of Warriner, Adelman, Aryani, de Zubicaray, and the OSS community.

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## **A Per-feature results**

The complete table of all 68 features  $\times$  Pearson  $r$  with valence on the 13,914 Warriner lemmas is in `results/word_level_correlations.csv`.

## **B Feature engineering details**

The full implementation of the feature extractor is in `src/features.py`. Unit tests covering 33 hand-computed expected values are in `tests/test_features.py`.